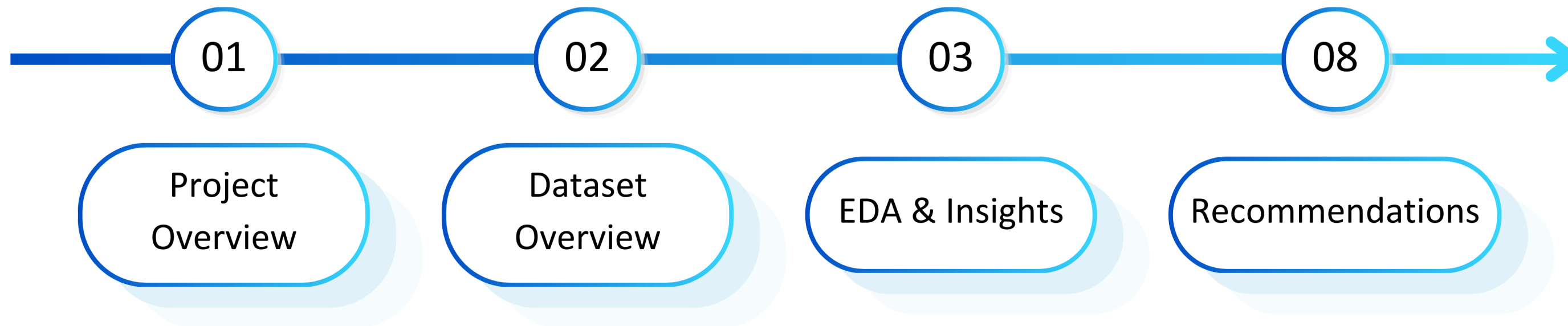


Data Analyst Project

Job Vacancy Tweets Analysis

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Understanding the business problem, objectives, and analytical framework

Business Understanding

Twitter has become a popular platform for companies to share job vacancies and reach potential candidates.

Analyzing companies, job positions, engagement, and posting patterns helps uncover hiring trends in the digital labor market.

Objectives

Identify the most active users based on posting volume and engagement during the observation period.

Analyze job posting behavior across year, day of week, and hour to uncover productivity patterns and engagement timing.

Identify the most frequently advertised job positions and their engagement levels.

Framework

Data Collection

Obtain the Job Vacancy Tweets dataset from Kaggle and explored 50,000 rows × 13 columns.

Data Preparation

Process, transform, clean, and standardize the raw dataset into an analysis-ready format.

Exploratory Data Analysis

Identify hiring trends, posting behavior, and engagement patterns in job vacancy tweets.

Dashboard Development

Build an interactive dashboard in Google Data Studio.

Insights Generation

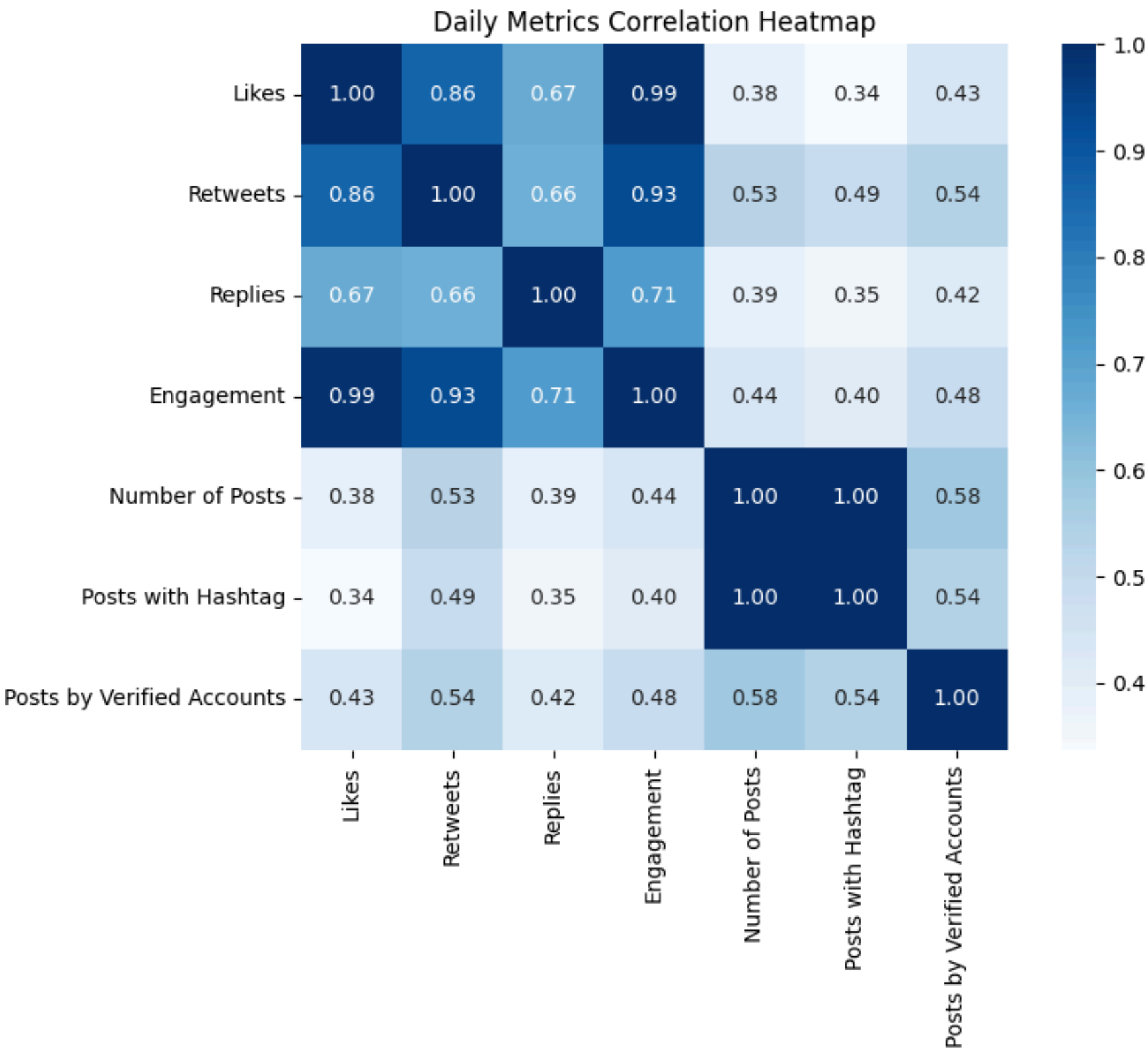


Dataset Overview

Overview dataset structure, quality, and variables

Dataset Size	Period	Variables	Description
50,000 × 13	2019–2023	ID	The unique identifier of the tweet
		Timestamp	The date and time when the tweet was posted
Total Missing Values	Total Duplicate Rows	User	The Twitter handle of the user who posted the tweet
15,305	0	Text	The content of the tweet
		Hashtag	The hashtags included in the tweet, if any
Source	Author	Retweets	The number of retweets as of the time it was scraped
<u>Kaggle</u>	Prasad Patil	Likes	The number of likes as of the time it was scraped
		Replies	The number of replies as of the time it was scraped
		Source	The source application or device used to post the tweet
		Location	The location listed on the user's Twitter profile, if any
		Verified_Account	A Boolean value whether the user's account has been verified
		Followers	The number of followers as of the time the tweet was scraped
		Following	The number of following as of the time the tweet was scraped
Additional Notes			
- Engagement is sum of Retweets, Likes, and Replies. - Location isn't used in this analysis because it's highly inconsistent and cannot reliably represent users' actual geographic locations. - NaN data from hashtag column will not be removed.			

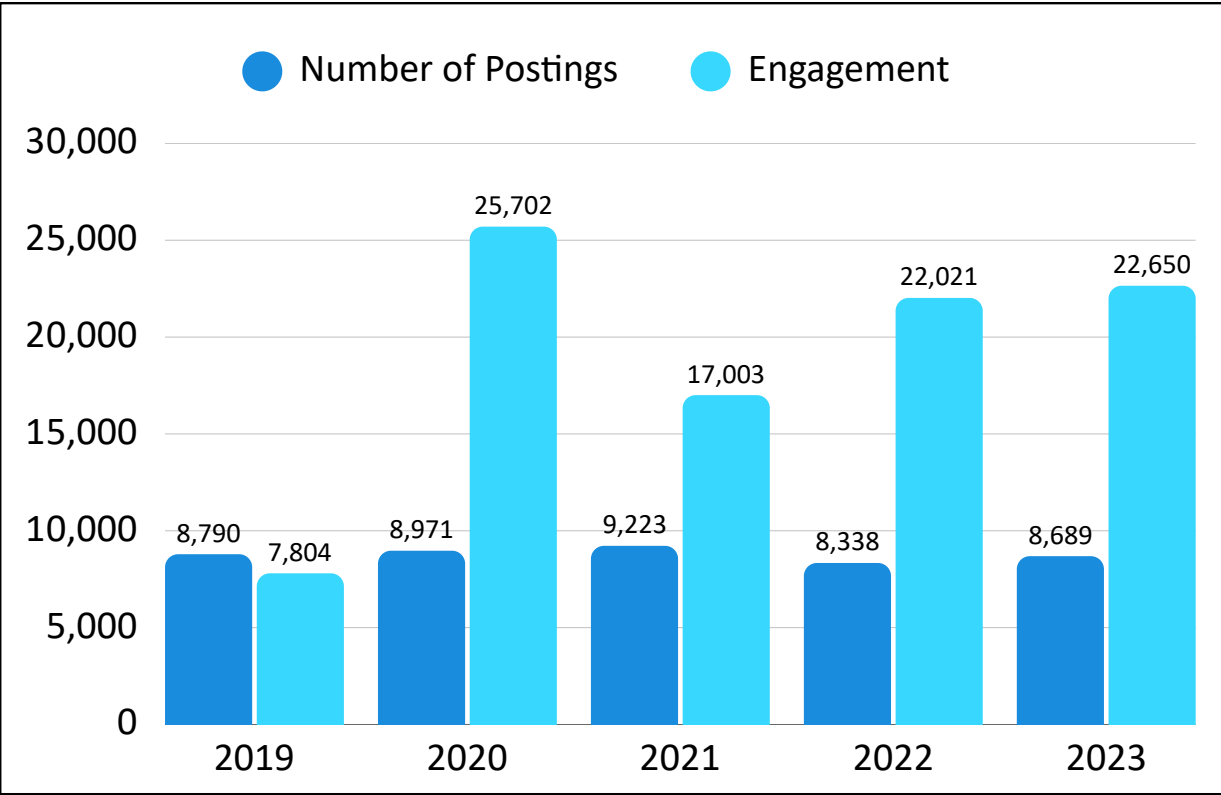
Heatmap Correlation



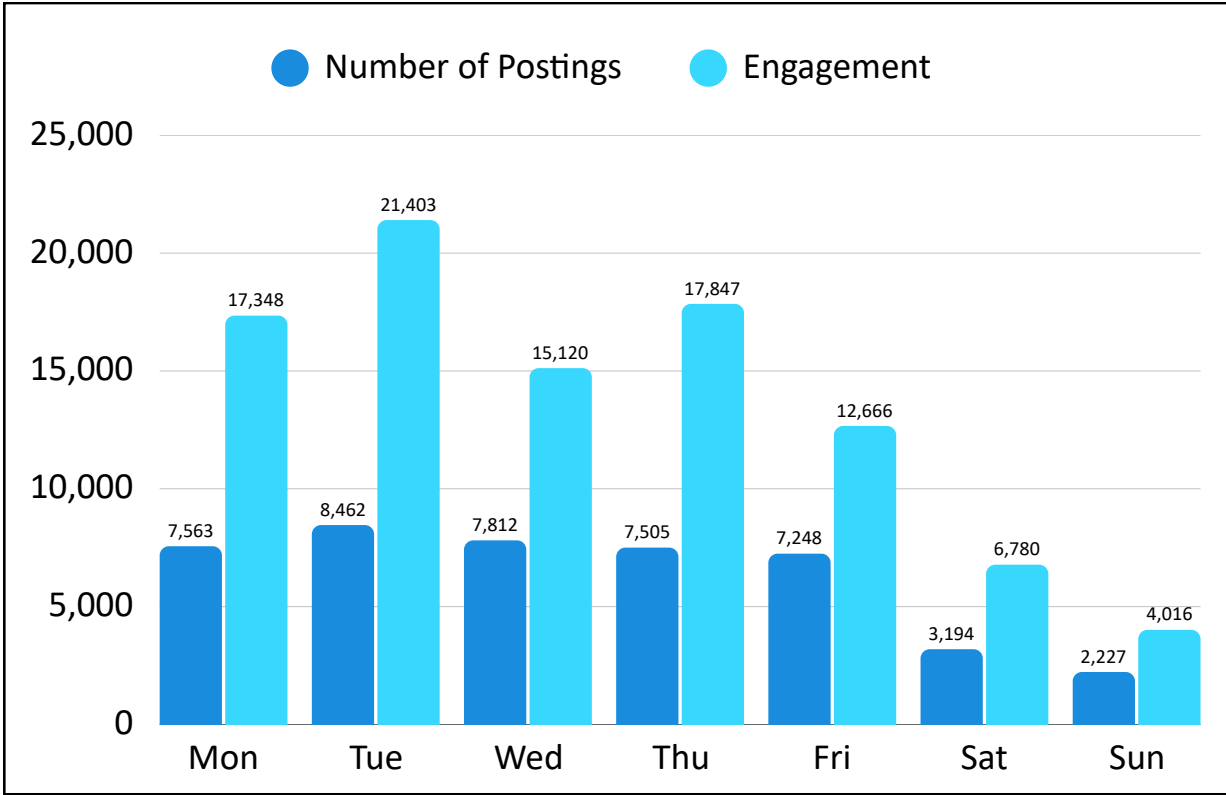
Insights

- Posting volume has a moderate correlation with engagement ($r = 0.44$), suggesting that **content quality and relevance matter more than posting frequency**.
- Verified accounts show a stronger correlation with engagement ($r = 0.48$) than posting volume, highlighting the **importance of account credibility**.
- Hashtag usage has a relatively weak correlation with engagement ($r = 0.40$), indicating **it supports discoverability but isn't a key engagement driver**.

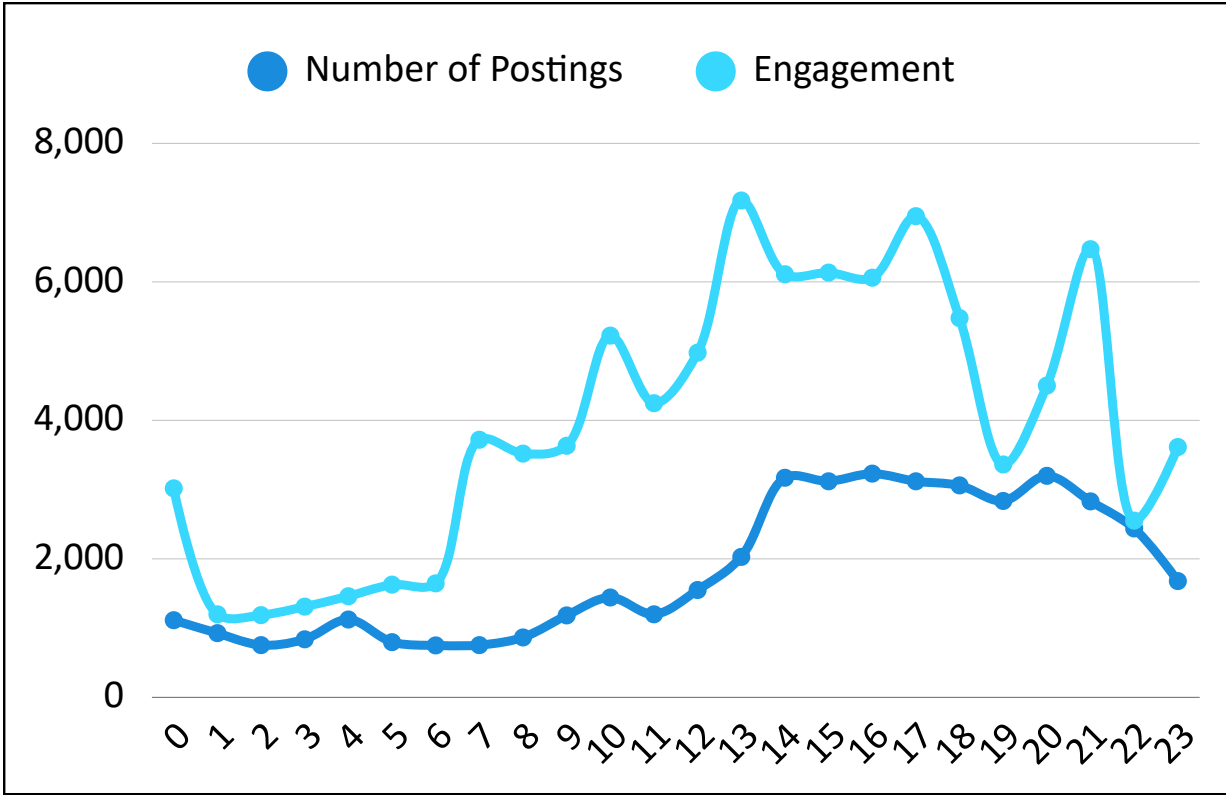
Number of Postings per Year



Number of Postings per Day of the Week



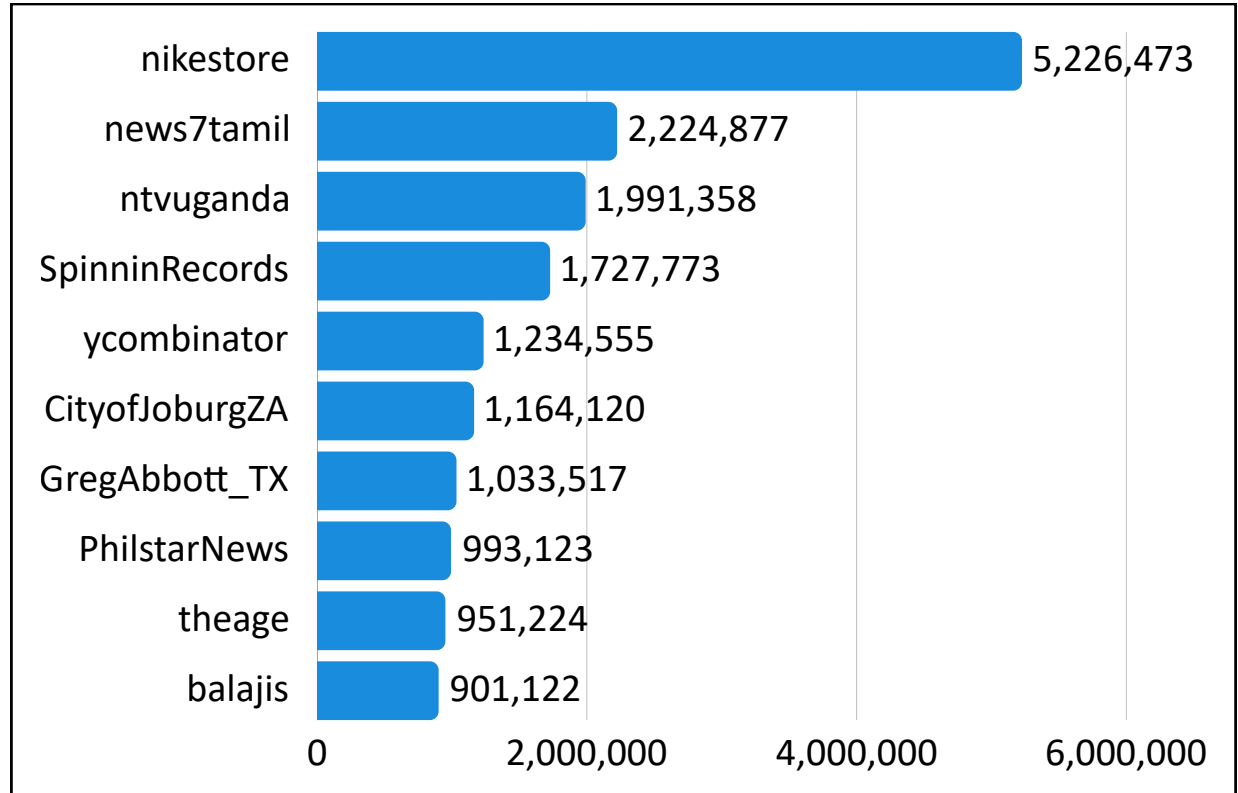
Number of Postings per Hour



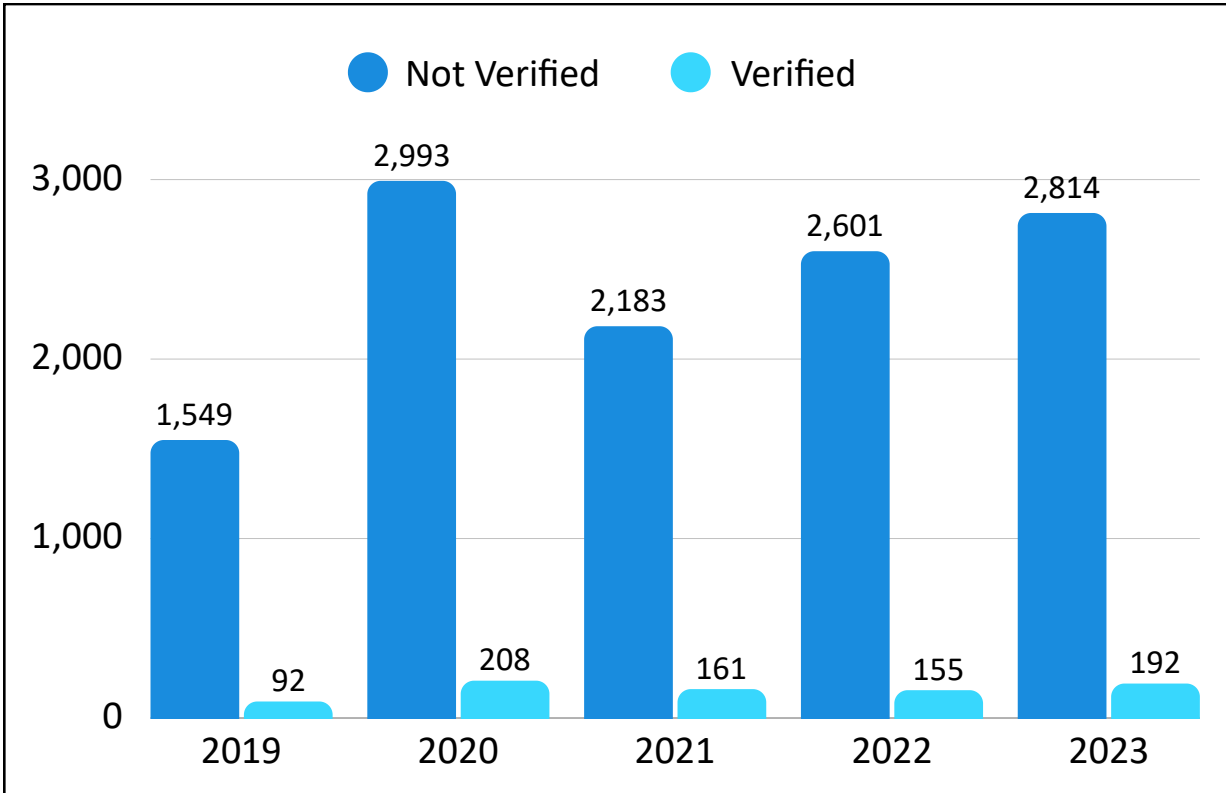
Insights

- Posting activity remained relatively stable (± 8.8 tweets/year). Engagement fluctuated significantly, indicating **engagement is influenced more by content performance than posting volume**.
- **Weekday posts consistently outperformed weekends**, with Tuesday generating the highest engagement ($\pm 21.4K$), while weekends recorded the lowest posting activity and engagement.
- **Posting volume was concentrated between 14:00–21:00**, suggesting this is the primary publishing window for discussions and content distribution.
- **Engagement efficiency concentrated between 13:00–17:00 and at 21:00**, indicating stronger audience responsiveness during these periods.

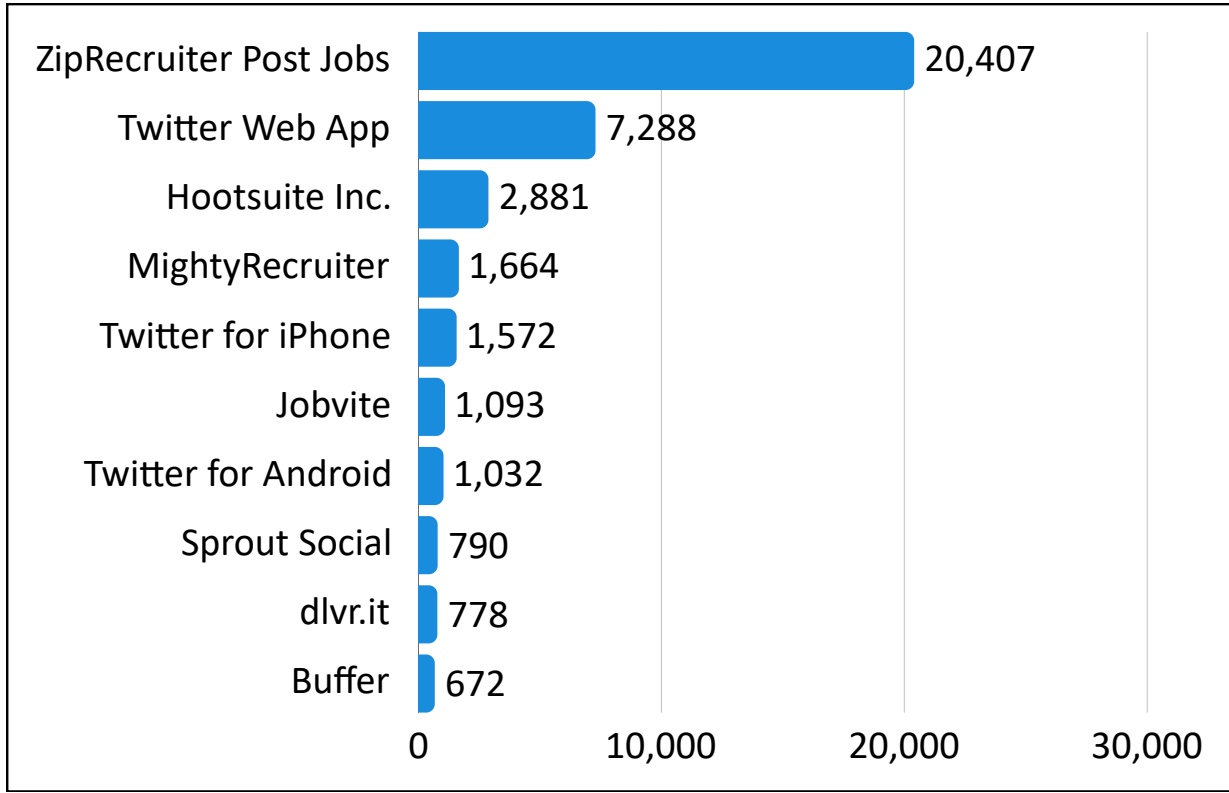
Top 10 User by Followers



Number of Unique Users by Account Status



Top 10 Sources



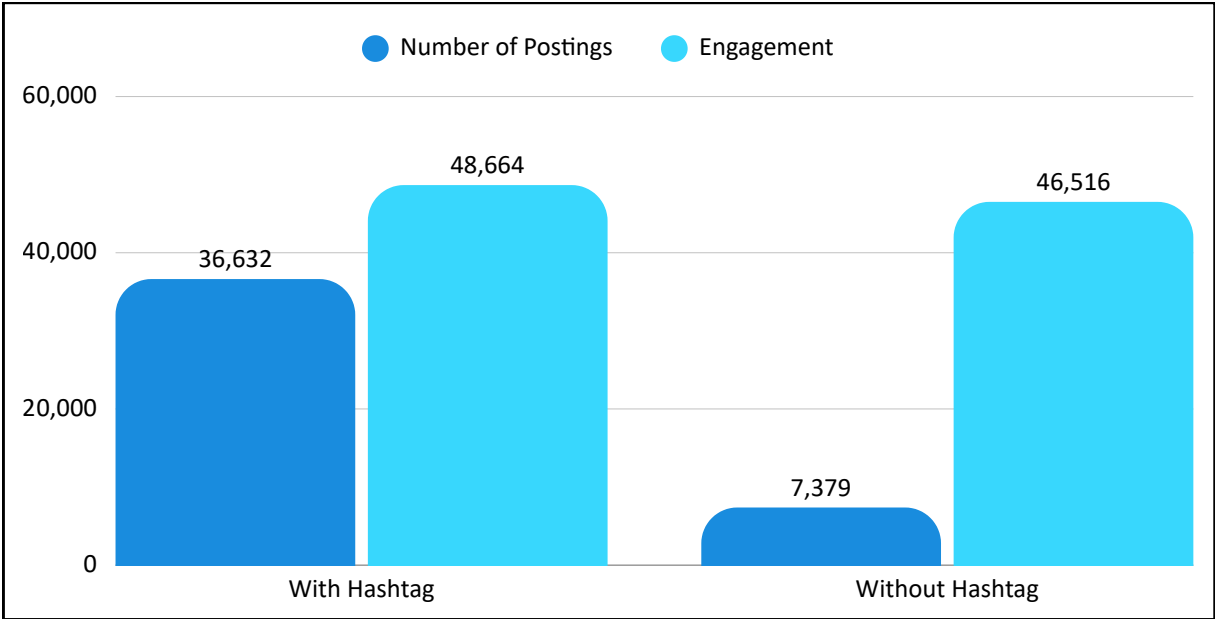
Insights

- **Conversation volume was dominated by non-verified accounts across all years ($\pm 93.8\%$),** indicating discussions were primarily driven by the broader user community.
- **Job vacancies were primarily posted by high-reach accounts,** with several recruiters, media outlets, and organizations having follower bases exceeding 900K, helping expand visibility.
- **Posting activity was heavily driven by third-party publishing tools,** indicating a strong reliance on automated content distribution rather than purely organic posting.

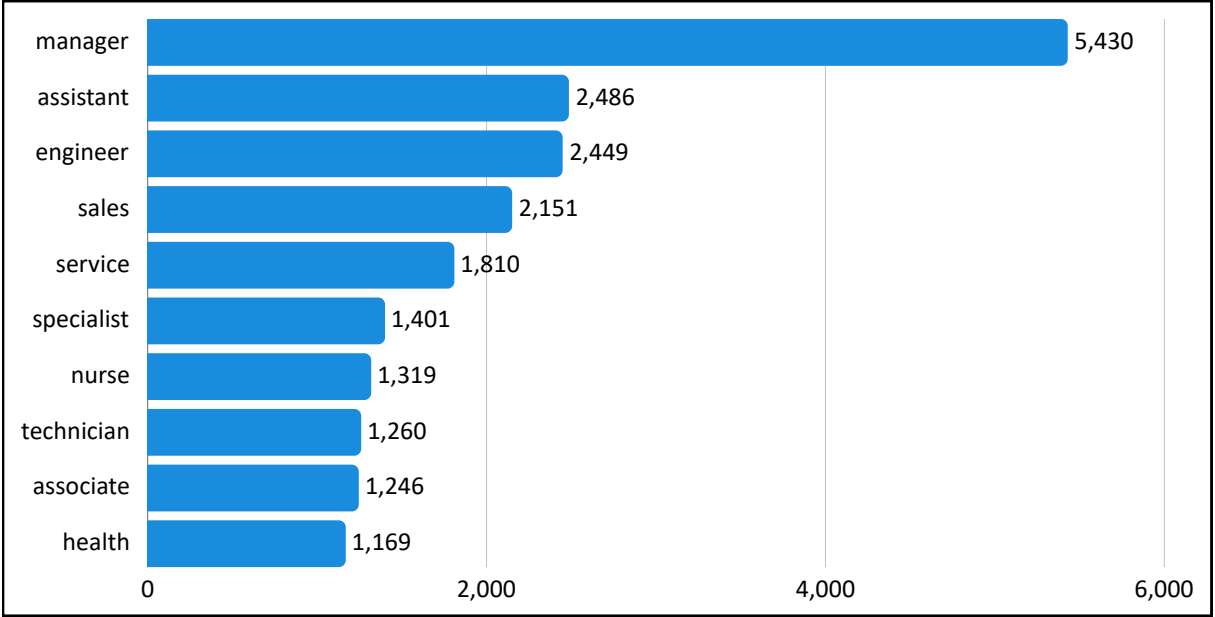
Top 50 WordCloud



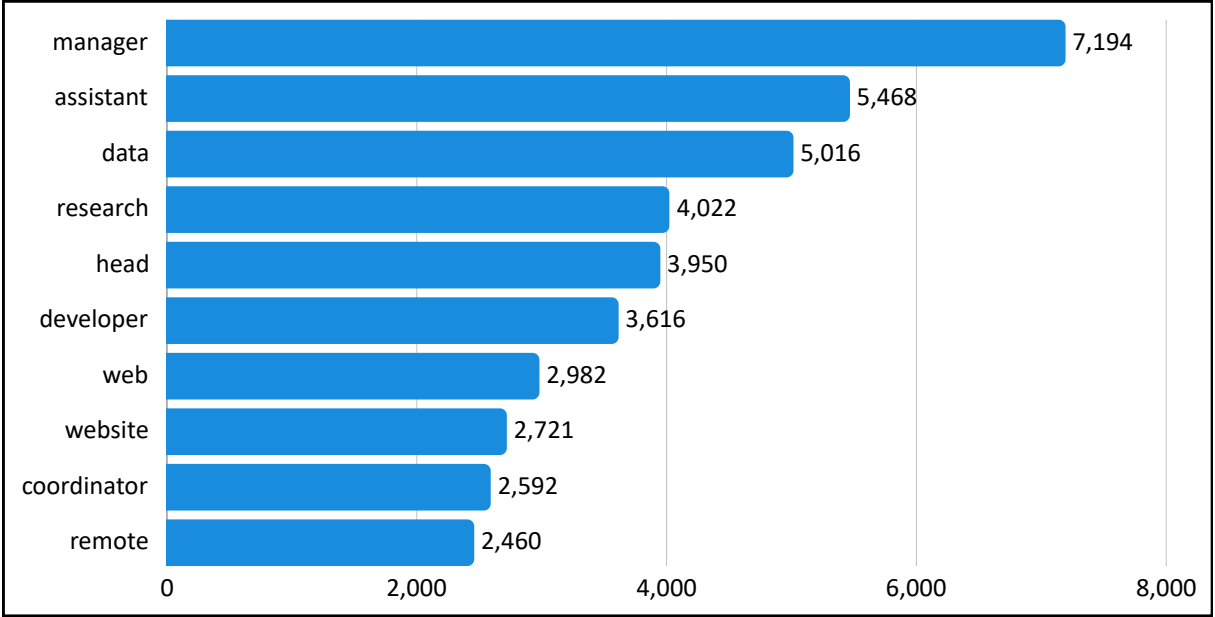
Hashtag Analysis



Top 10 The Most Looking Job and Field

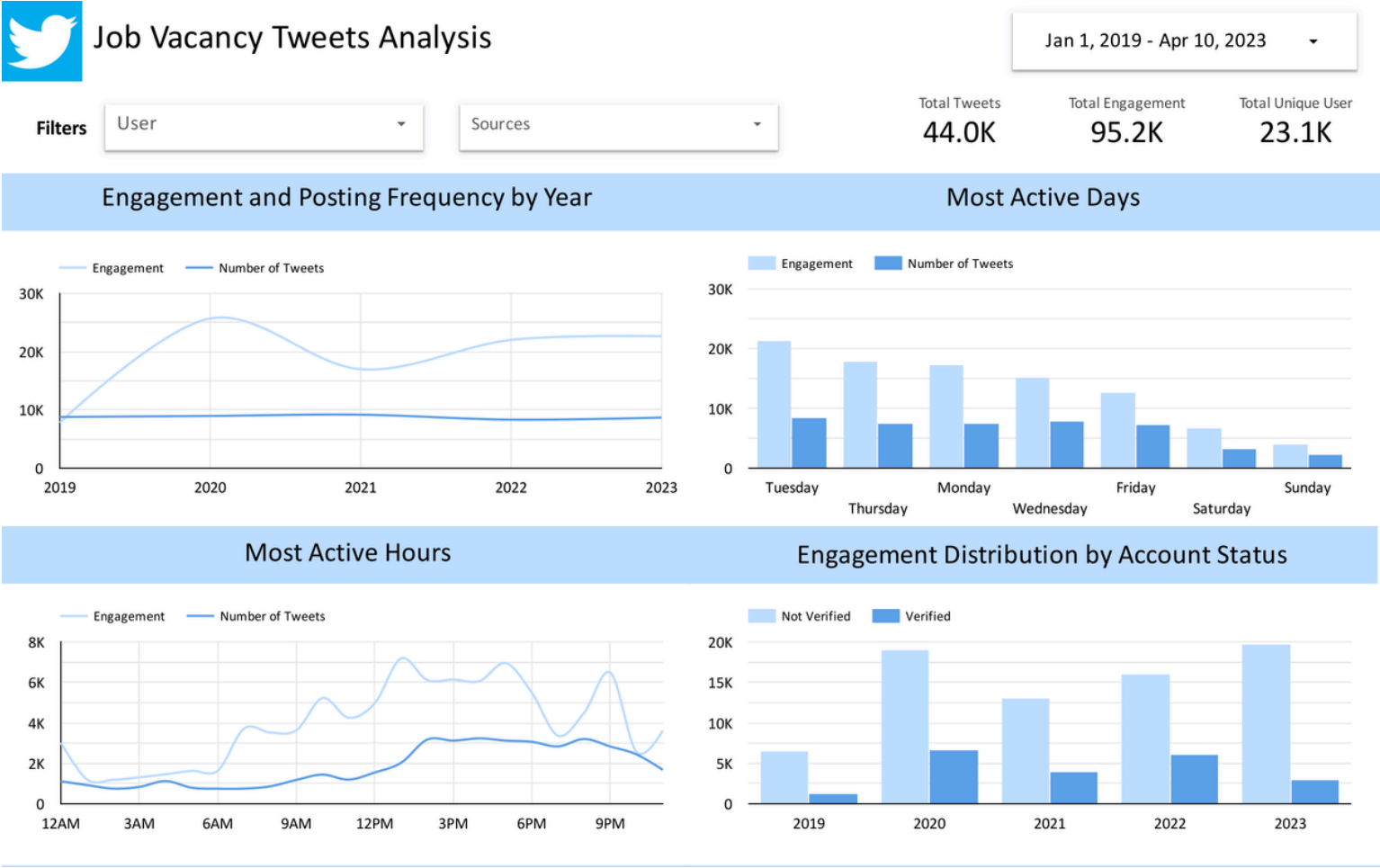


Top 10 Job and Field by Engagement

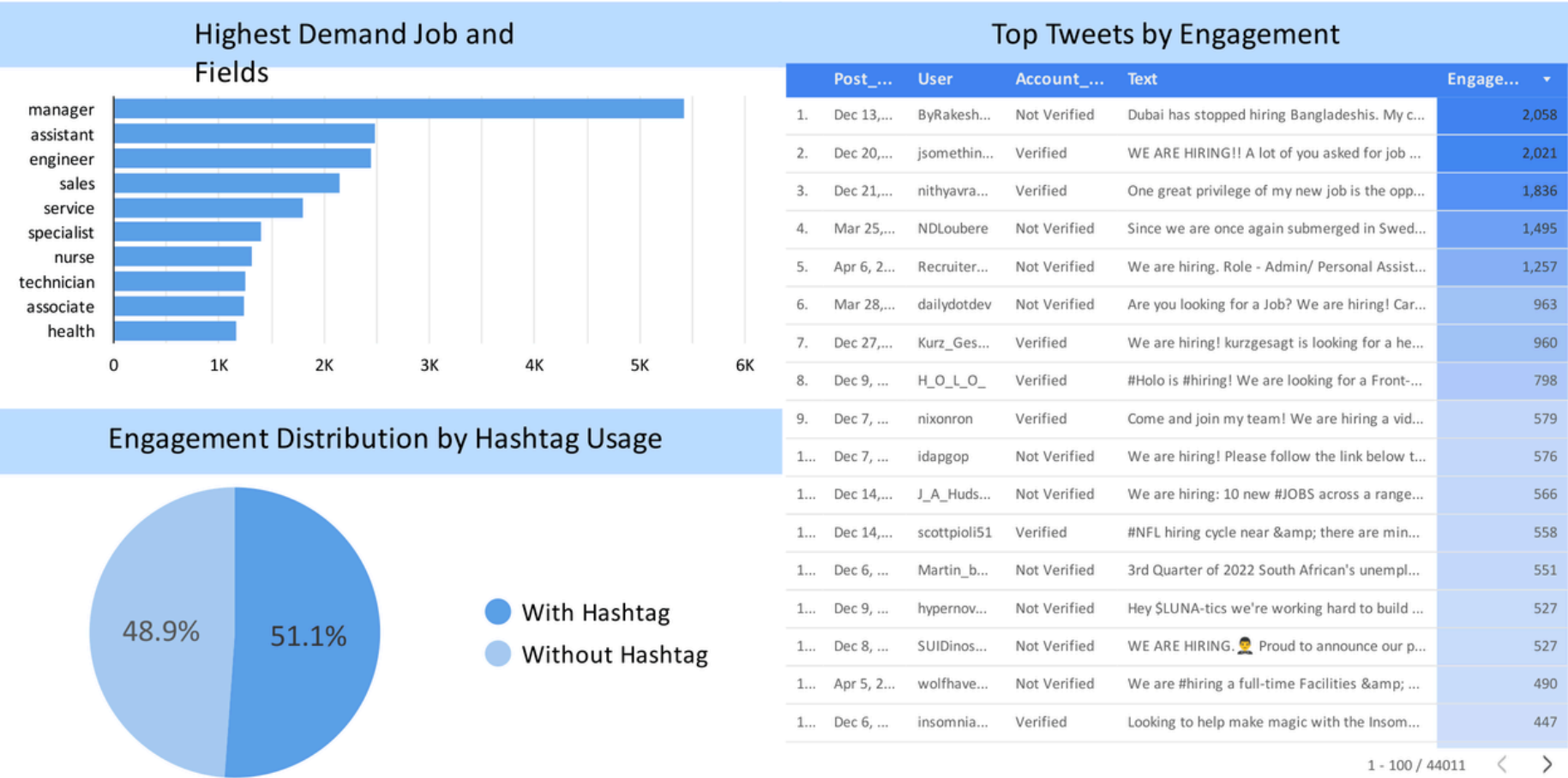


Insights

- Tweets with hashtags slightly outperformed tweets without them in engagement, indicating **hashtags are one of a key engagement driver**.
- Manager and assistant were the most frequently mentioned and engaged roles in job vacancy tweets, showing **strong demand for this positions**.



Link Dashboard



Recommendations	Actionable Insights	Expected Impact
Publish During Prime Day and Time	Prioritize posting on Tuesday–Thursday and active working time, especially at 13:00–17:00.	Increase tweet visibility and median engagement by reaching the audience when they are most active.
Increase the Use of Hashtag	Increase the use of 1–3 hashtag in tweets, such as #Hiring, #NameoftheJob, #NameoftheCompany.	Improve content discoverability and organic reach, attracting a broader audience and increasing engagement.
Focus on Quality Content	Focus on content quality and audience relevance rather than simply increasing posting volume.	Improve engagement efficiency by generating higher interactions per tweet.
Focus on High Demand Job	Develop more content around high-engagement job categories, such as manager and data, research, and developer roles.	Increase engagement by publishing content that better matches audience demand, resulting in more interactions and potential applicants.

LET'S CONNECT



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